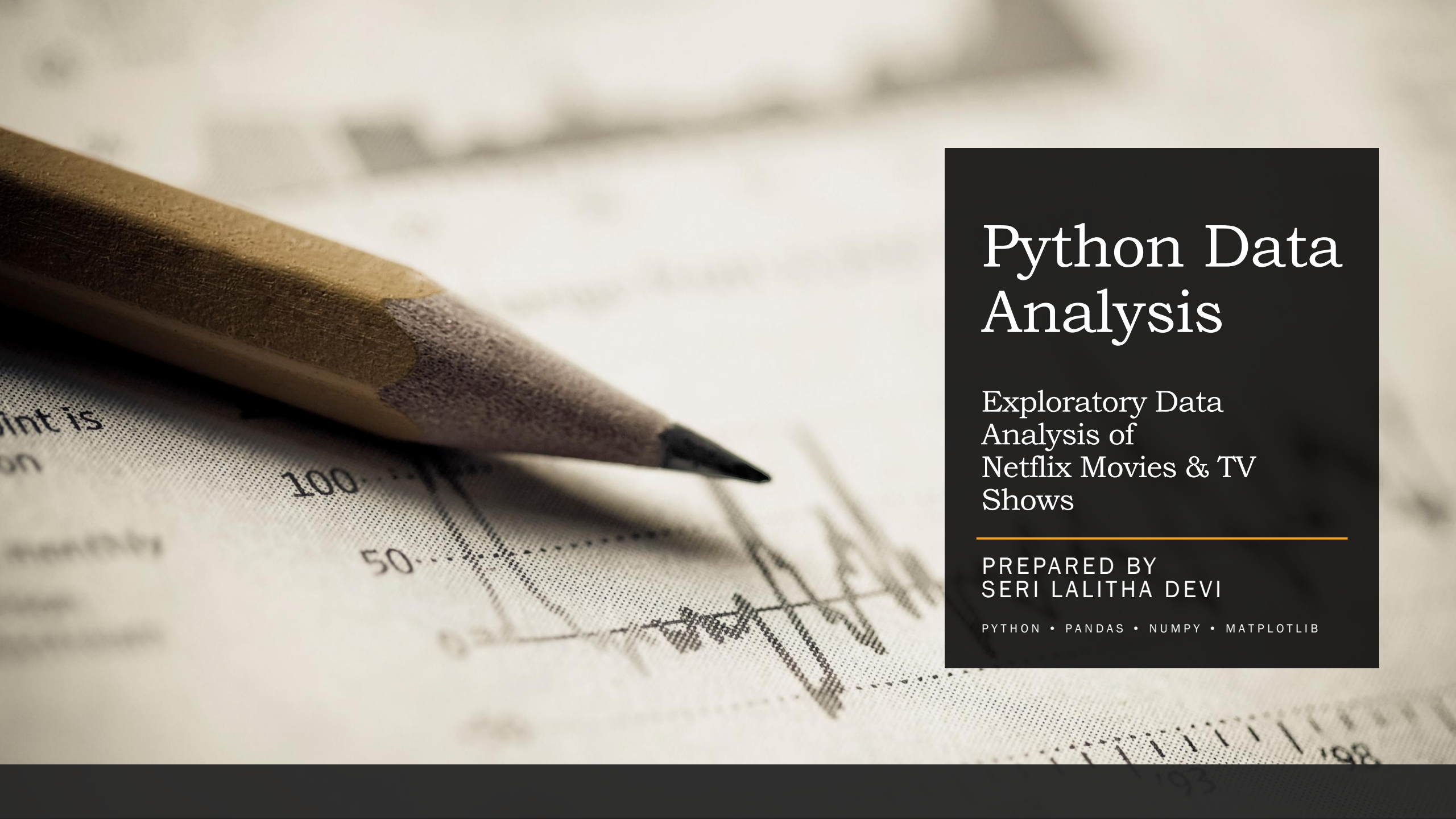




Python Data Analysis

Exploratory Data
Analysis of
Netflix Movies & TV
Shows

PREPARED BY
SERI LALITHA DEVI

PYTHON • PANDAS • NUMPY • MATPLOTLIB

Project Overview

Netflix is one of the world's leading streaming platforms, offering a diverse collection of movies and TV shows across multiple countries and genres. Analyzing its content library provides valuable insights into content distribution, production trends, and audience preferences.

This project performs **Exploratory Data Analysis (EDA)** on the Netflix Movies & TV Shows dataset using Python. The analysis includes data cleaning, visualization, and insight generation to identify trends in content type, release patterns, ratings, genres, and geographic distribution.

Project Scope

- Analyze content distribution
- Examine release trends
- Identify top countries & genres
- Explore content ratings
- Generate business insights

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Jupyter Notebook

Business Objective

This project aims to analyze Netflix's content library and answer key business questions using data analysis and visualization techniques.

Objectives

- Identify the distribution of Movies and TV Shows.
- Analyze content growth over time.
- Discover the top content-producing countries.
- Examine ratings and audience classifications.
- Identify the most common genres.
- Generate actionable business insights from the data.

Expected Outcome

The analysis helps understand Netflix's content strategy, audience targeting, and content trends, enabling data-driven business decisions.

Dataset Overview

The analysis is based on the **Netflix Movies & TV Shows** dataset, which contains detailed information about movies and TV shows available on Netflix. The dataset provides metadata that helps analyze content distribution, release patterns, genres, ratings, and geographic trends.

Dataset Information

- Source:** Kaggle – Netflix Movies & TV Shows Dataset
- Records:** 8,807
- Features:** 12
- Format:** CSV

Key Attributes

Show ID, Title, Type, Director, Cast, Country, Date Added, Release Year, Rating, Duration, Genre & Description.

Metric	Value
Total Records	8,807
Total Features	12
File Format	CSV
Data Source	Kaggle

Data Cleaning

Data cleaning was performed to improve the quality and reliability of the dataset before analysis.

Data Cleaning Process

- Checked dataset structure and data types.
- Identified missing values across columns.
- Removed duplicate records.
- Converted the **Date Added** column to datetime format.
- Verified data consistency for analysis.

Outcome

- Clean and structured dataset.
- Accurate date formatting for time-based analysis.
- Improved data quality for visualization and business insights.
- Dataset prepared for Exploratory Data Analysis (EDA).

```
[10]: df.isnull().sum().sort_values(ascending=False)
```

```
[10]: director      2634  
country      831  
cast         825  
date_added    10  
rating        4  
duration      3  
show_id       0  
type          0  
title         0  
release_year  0  
listed_in     0  
description    0  
dtype: int64
```

```
[11]: print("Duplicate Rows :", df.duplicated().sum())  
Duplicate Rows : 0
```

```
[5]: df.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 8807 entries, 0 to 8806  
Data columns (total 12 columns):  
#   Column          Non-Null Count  Dtype  
---  ---  
0   show_id         8807 non-null   str  
1   type            8807 non-null   str  
2   title           8807 non-null   str  
3   director        6173 non-null   str  
4   cast            7982 non-null   str  
5   country         7976 non-null   str  
6   date_added      8797 non-null   str  
7   release_year    8807 non-null   int64  
8   rating          8803 non-null   str  
9   duration        8804 non-null   str  
10  listed_in       8807 non-null   str  
11  description      8807 non-null   str  
dtypes: int64(1), str(11)  
memory usage: 447.3 KB
```

Exploratory Data Analysis (EDA)

EDA was conducted to identify trends, patterns, and relationships within Netflix's content library through data visualization.

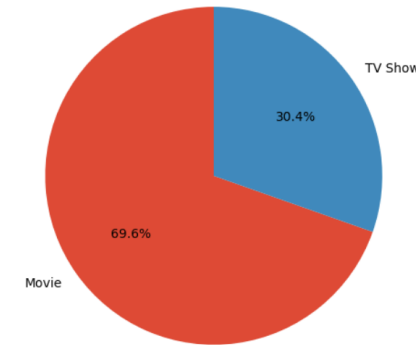
Visualizations Performed

- Content Distribution (Movies vs TV Shows)
- Top 10 Content-Producing Countries
- Content Ratings Distribution
- Content Added Over Time
- Release Year Trend
- Top 10 Genres
- Top 10 Directors
- Movie Duration Distribution
- Top Release Years
- Content Type Distribution

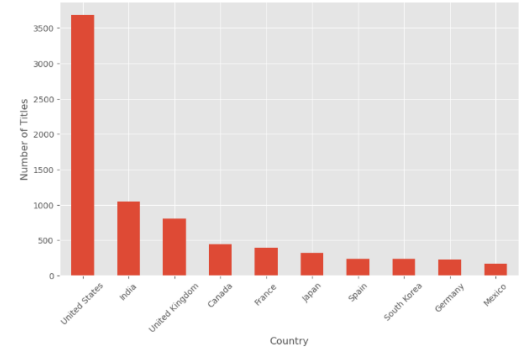
Key Focus Areas

Content Type • Countries • Ratings • Release Trends • Genres • Directors

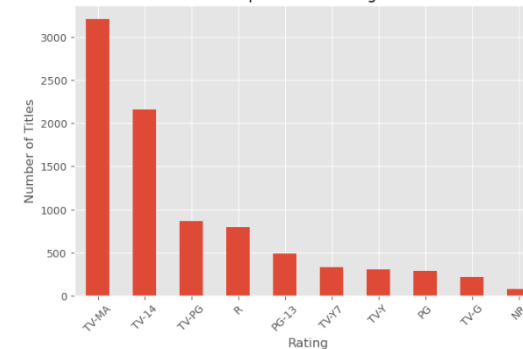
Movies vs TV Shows



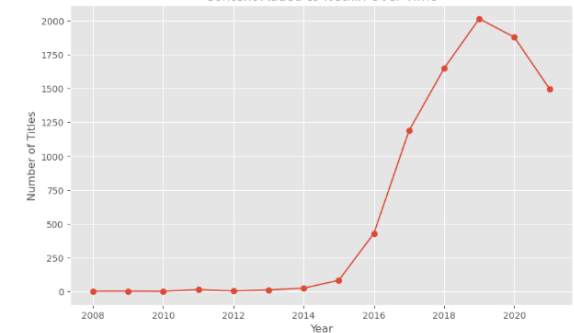
Top 10 Countries by Content



Top Content Ratings



Content Added to Netflix Over Time



Key Insights

Based on the exploratory data analysis, the following insights were identified:

- **Movies** make up the majority of Netflix's content library.
- The **United States** contributes the highest number of titles.
- **TV-MA** is the most common content rating.
- Netflix's content library experienced significant growth **after 2015**.
- **International Movies** and **Dramas** are among the most popular genres.
- Most movies have a duration between **80–120 minutes**.

Recommendations

Based on the analysis, the following recommendations can help strengthen Netflix's content strategy:

- Expand investment in **international content** to reach a broader audience.
- Increase production in **high-performing genres** such as Drama and International Movies.
- Maintain a balanced mix of **Movies** and **TV Shows** to cater to diverse viewer preferences.
- Focus on **data-driven content acquisition** by analyzing audience demand and viewing trends.
- Continuously monitor **content ratings** and **release trends** to optimize future content planning.

Conclusion

This project successfully analyzed the **Netflix Movies & TV Shows** dataset using Python to uncover meaningful trends in content distribution, genres, ratings, release patterns, and geographic presence.

The analysis demonstrated the complete data analysis workflow, including data cleaning, exploratory data analysis (EDA), visualization, and business insight generation. The findings highlight how Python can be used to transform raw data into actionable insights that support data-driven decision-making.

Thank You

Thank you for your time and attention.

I appreciate the opportunity to present this project and demonstrate my Python data analysis skills. I look forward to applying these analytical techniques to solve real-world business problems.

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